

Supplementary Materials

How does familiarity impact the computational processes that support face recognition?

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A1 Appendix A: Supplementary Materials

A1.1 Parameter recovery study

A1.2 Priors

These are the priors used for our pre-registered models. The example parameter names here are for Experiments 1 and 2 (i.e., Novel vs Learned), but these priors were used in all experiments.

parameter	description	condition	correct	response	mean	sd
v_SNovel:IMFALSE	Drift rate	Novel	False	-	0	2
v_SLearned:IMFALSE	Drift rate	Learned	False	-	0	2
v_SNovel:IMTRUE	Drift rate	Novel	True	-	1	2
v_SLearned:IMTRUE	Drift rate	Learned	True	-	1	2
sv_IMTRUE	SD drift rate (true accumulator)	-	True	-	$\log(0.5)$	0.5
B_IRNovel	Threshold	-	-	Novel	$\log(1)$	0.5
B_IRLearned	Threshold	-	-	Learned	$\log(1)$	0.5
A	Start point	-	-	-	$\log(0.5)$	0.3
t0	Non-decision time	-	-	-	$\log(0.3)$	0.3

A1.3 Model checks

A1.3.1 Convergence

The chains showed satisfactory mixing as evidenced by the rhat values.

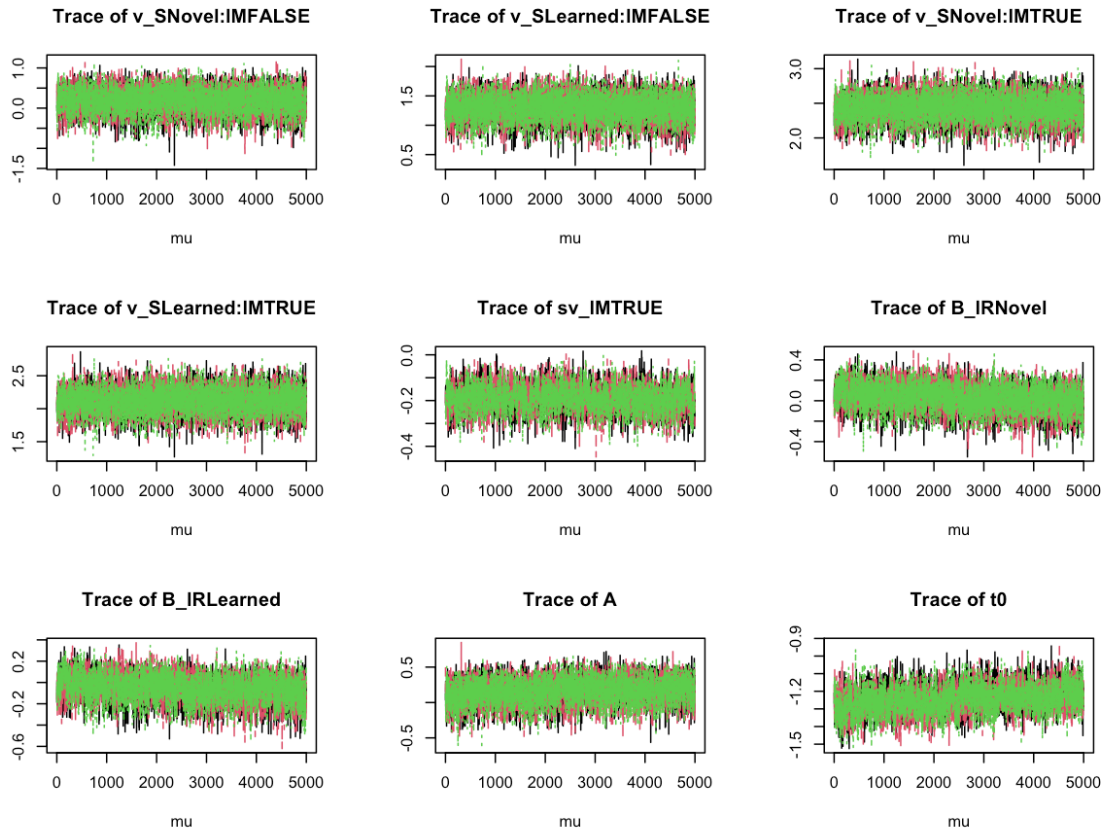
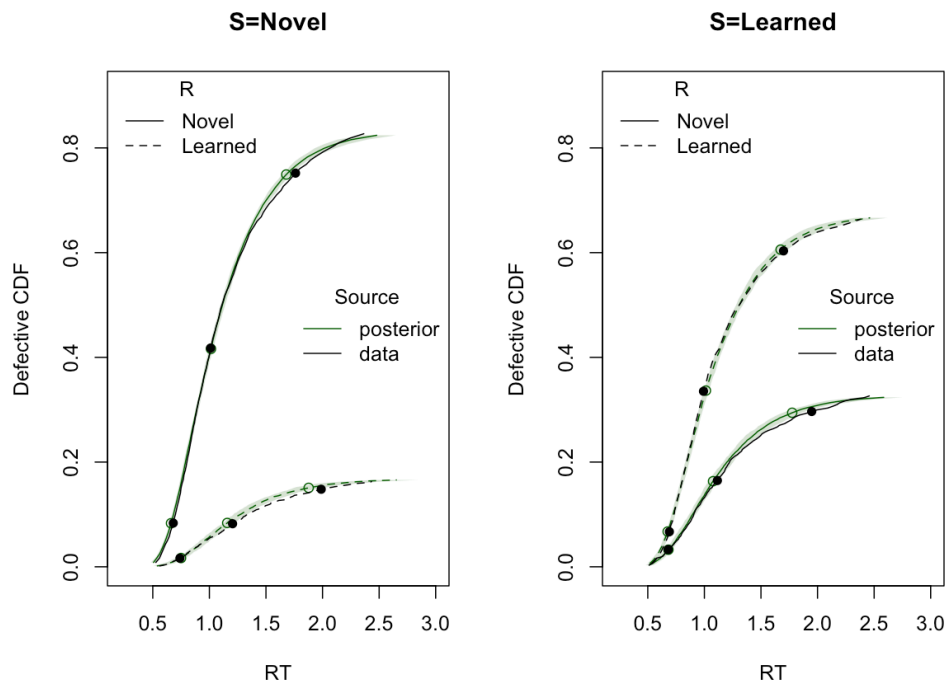
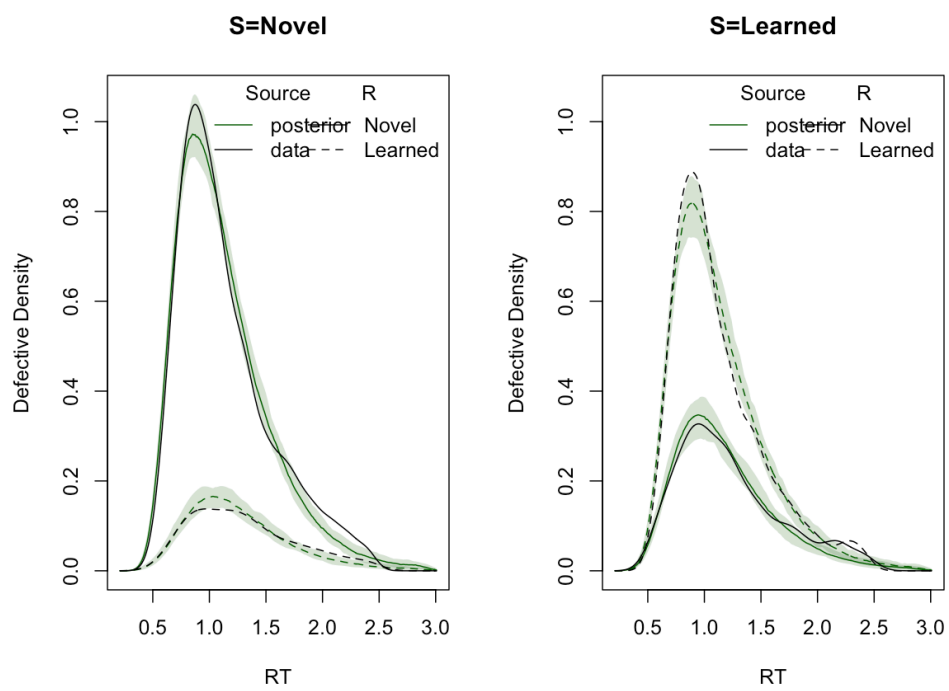


Figure A1.1. MCMC chains for Experiment 1

A1.3.2 Posterior predictive checks



(a) CDF



(b) Density

Figure A1.2. Posterior predictive checks for Experiment 1

A1.3.3 Prior vs posterior plots

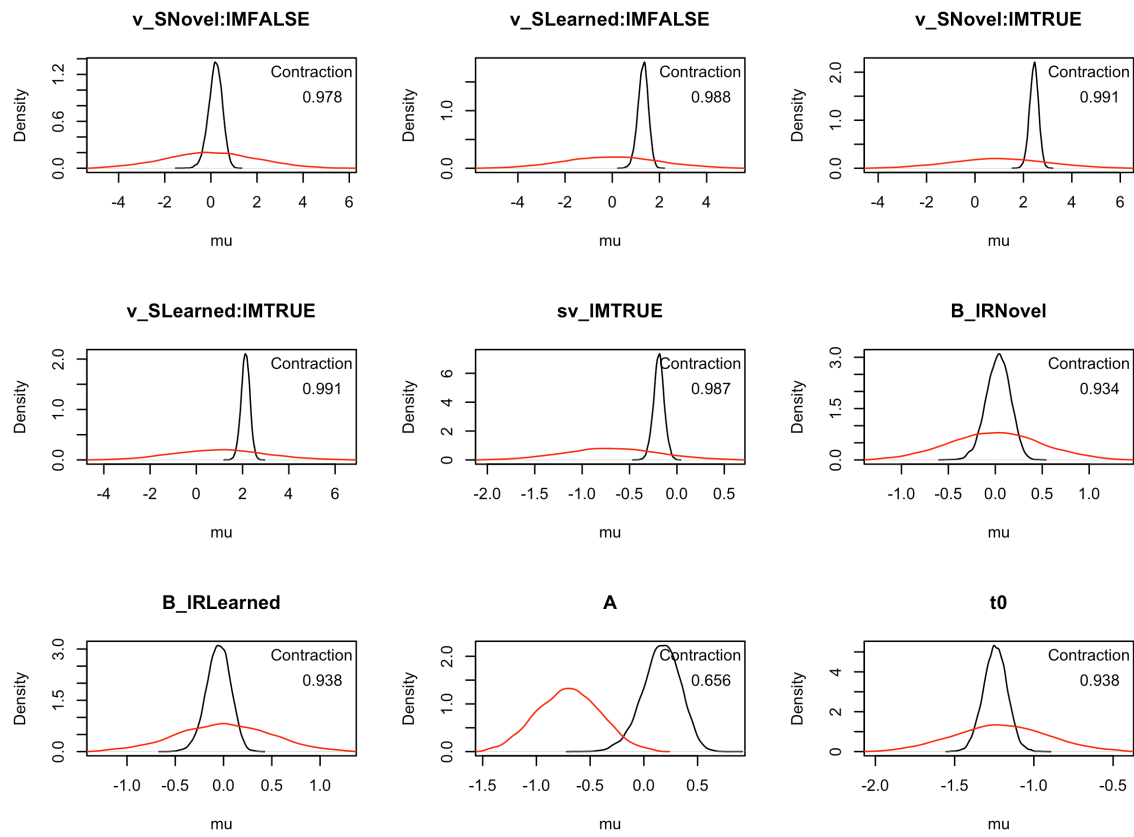


Figure A1.3. Priors vs posterior plots for Experiment 1

A1.4 Exploratory models

References